

TITLE: PlumPay LabFrame V1: Behavioral Spoof Simulations of Entropy-Based Identity Systems Without Operational Transmission

SUBTITLE: A Local Simulation Protocol for Predictive Deception Training in Cryptographic Authentication Frameworks

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ABSTRACT:

This whitepaper introduces a non-operational cryptographic simulation protocol — PlumPay LabFrame V1 — designed to explore

SECTION 1: BACKGROUND & MOTIVATION

Modern financial authentication mechanisms — such as Apple Pay — rely on a Secure Element (SE) to generate transaction-specific

PlumPay was developed under this new lens. The hypothesis: if a system accepts a payload based not on perfect cryptographic t

SECTION 2: SYSTEM OVERVIEW

PlumPay simulates:

- Phantom BLE broadcast toggling (no RF emissions)
- UID drift emulation and echo response logging
- Entropy modulation based on local density conditions
- Tactical collapse fallbacks for system retraction
- Synthetic cryptograms using OpenSSL hash digests
- Logging microdrift UID deltas
- Predicting nonce-like payloads reversed from entropy
- NFC payload injection emulation (non-operational)

All actions are strictly confined to terminal-based logs and shell calls, operating in a fully airgapped, non-transmissive framework.

SECTION 3: RESULTS FROM SIMULATION LOGS

Over several test cycles, the framework was observed to:

- Generate UID drift patterns mimicking legitimate delta noise
- Construct synthetic payloads aligning with expected cryptogram timing
- Simulate terminal responses resembling backend fallback behavior

In no instance was a real NFC transmission, BLE packet emission, or secure payment interaction performed. All simulations were

SECTION 4: IMPLICATIONS

These results suggest a novel attack simulation method: narrative spoofing of cryptographic timing, where the illusion of a valid tra

SECTION 5: ETHICAL LIMITATIONS

This research was conducted in a fully offline, simulated context. No encryption keys were derived, transmitted, or stored. No cont

The authors acknowledge that any attempt to move beyond simulation — especially into real NFC hardware injection or packet re

SECTION 6: PROPOSED CLASSIFICATION LEVEL

Suggested Tier: SAP-compatible / Restricted (black-box simulation only)

Recommended Use: Strategic training, adversarial modeling, environmental deception simulation

Redaction Level: All output logs are anonymized, with no cryptographically actionable material

SECTION 7: CONCLUSION & FUTURE WORK

The PlumPay project is an invitation: not to break encryption, but to predict the illusion of legitimacy. This work challenges how we

We propose continued sandboxed development for national training simulations, adversarial forensics, and behavioral anomaly te

This is not a blueprint for heists. It's a manual for the next generation of deception detection.

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